

Rockborne

Decisions Trees & Random Forest Classifiers

Many trees vote, and the crowd beats any single tree

Objectives



Intro to Random Forest

What a forest is and why many trees help.



Decision Trees

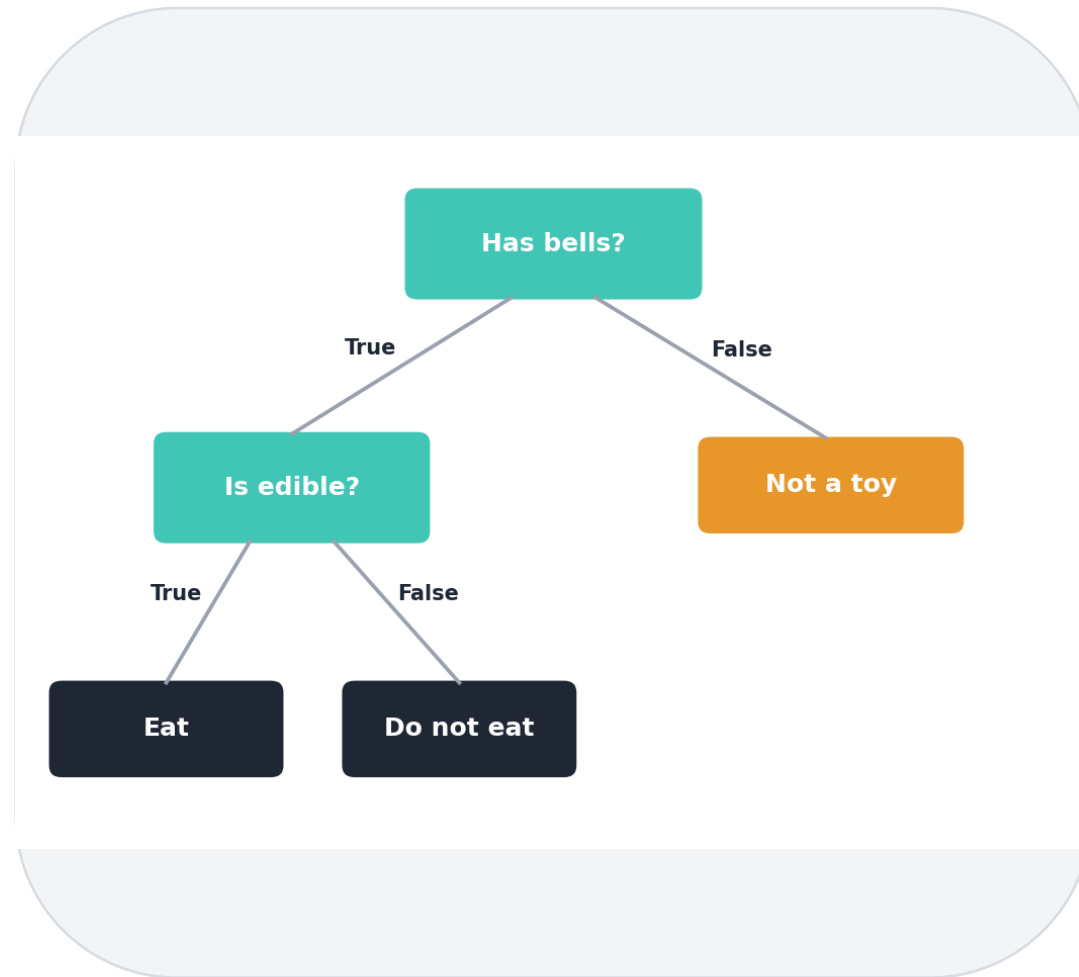
The building block: how a single tree decides.

Random Forest Classifier



Decision Trees

What is a Decision Tree?



1

A tree-shaped diagram that leads to a decision.

2

Each branch represents a possible choice or outcome.

3

Internal nodes ask a question; leaves give the answer.

4

Example: 'Has bells?' then 'Is edible?' to a final decision.

Decision Tree Terminology

Node types

- Root node: the topmost decision node.
- Decision node: a node with two or more branches.
- Leaf node: carries the final classification or decision.

Split measures

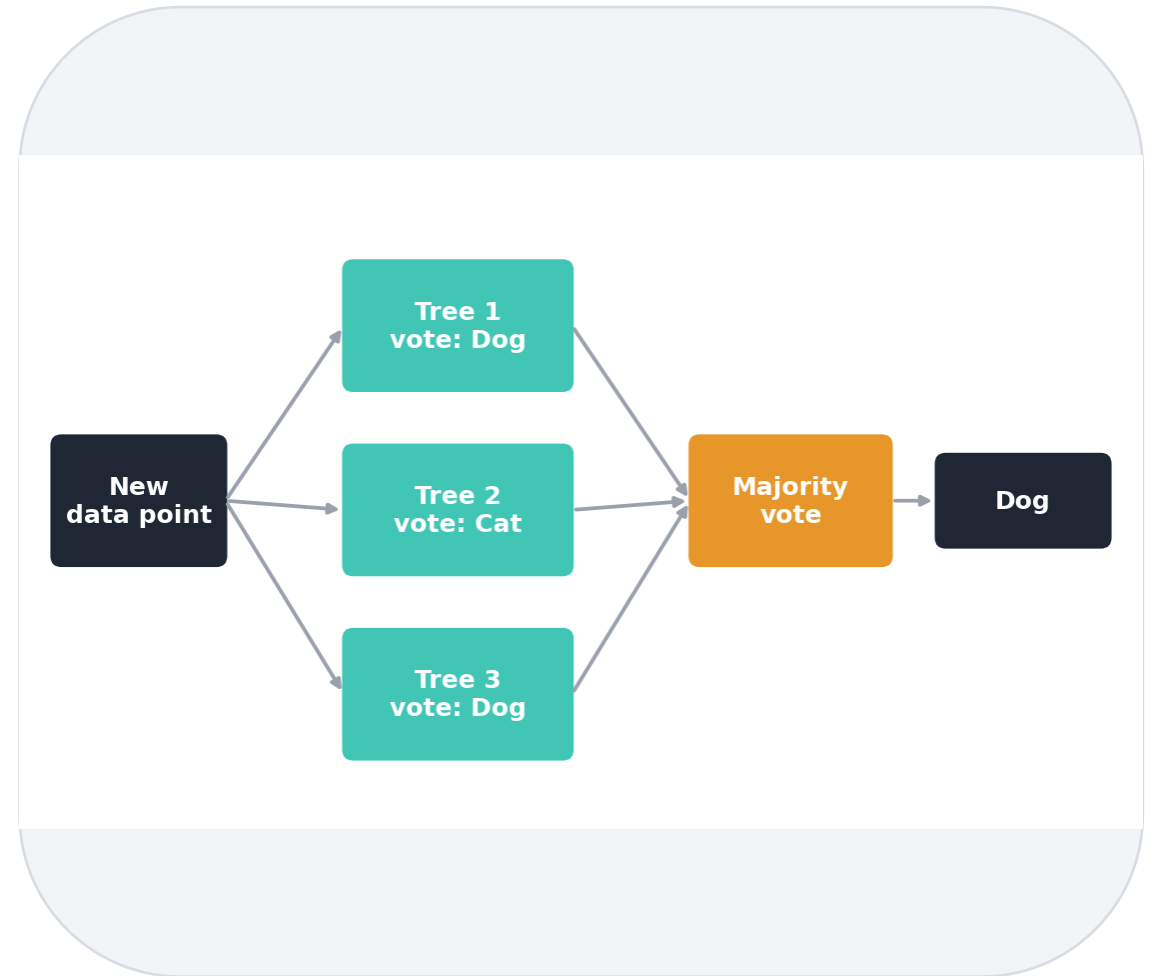
- Entropy: the measure of randomness in the dataset.
- Information gain: the drop in entropy after a split.
- Trees split on the feature with the highest information gain.

Introduction

Intro to Random Forest Classifier

What is a Random Forest?

- Builds many decision trees during training.
- Each tree makes its own decision, a vote.
- The majority vote across the trees is the final classification.
- Wisdom of the crowd: many trees beat one.



Where Random Forests are Used



Supervised learning

Classification and regression tasks.



Large datasets

Scales well to many rows and features.



Remote sensing

Classifying satellite and aerial imagery.



Object detection

For example, Microsoft Kinect body tracking.

Advantages and Disadvantages

Advantages

- Reduced risk of over-fitting (many trees).
- Generally very high accuracy.
- Efficient with large datasets; estimates missing data.
- Can be saved and reused; robust to noise.

Disadvantages

- Complex and hard for a human to interpret.
- Long training times.
- Larger models use more memory.

In this lesson, we explored



Random Forest

Many trees vote; the majority wins. High accuracy, less over-fitting.



Decision Trees

The building block: nodes, branches, leaves, entropy and information gain.

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Time to get hands-on in Python

We move from theory to practice with a guided walkthrough you run yourself in Python.



Decision Tree & Random Forest - Guided Walkthrough



Then: Independent Classification Labs

Digging Deeper?



[Random Forests](#)

Statology



[Decision Tree](#)

GeeksforGeeks



[Decision Tree Algorithm
Explained](#)

KDnuggets

Questions?